AI-Powered Personalized Travel Planning System: An Explainable Itinerary Generator Using Rule-Based Artificial Intelligence

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***Abstract*—Travel planning requires users to select destinations, organize daily activities, estimate expenses, and balance personal interests with available time and budget. Searching across multiple websites can be time-consuming, particularly for travellers unfamiliar with a destination. This paper presents an AI-Powered Personalized Travel Planning System that generates personalized day-wise itineraries using rule-based artificial intelligence. The system accepts user inputs such as destination, trip duration, budget, number of travellers, and interests including nature, adventure, history, culture, and shopping. A structured travel knowledge base stores attraction categories, estimated visit durations, entry fees, and approximate transport and food costs. A rule engine applies predefined conditions to select suitable attractions, organize activities, and identify budget-friendly alternatives. A budget estimation module calculates expected expenses and indicates when a plan may exceed the user's budget. The proposed system emphasizes explainability because recommendations can be traced to explicit rules and stored information. Unlike machine-learning-based approaches, the proposed system does not require model training or a large historical dataset. This paper describes the problem, related work, proposed methodology, architecture, implementation approach, testing plan, expected outcomes, limitations, and future enhancements. Experimental performance is not claimed because measured implementation results have not yet been provided.**

***Keywords*—Personalized travel planning, rule-based artificial intelligence, itinerary generation, budget estimation, recommendation system, explainable AI.**

I. INTRODUCTION

Travel planning involves several connected decisions, including selecting destinations, identifying tourist attractions, estimating travel time, choosing activities, and managing expenses. Travellers may have limited time and a fixed budget while also having different interests, such as nature, history, adventure, shopping, and cultural experiences. Manually combining these requirements can be time-consuming and may produce an itinerary that is expensive, unbalanced, or difficult to complete.

Traditional travel planning often depends on general travel websites, online reviews, and manually prepared schedules. Although these sources provide useful information, travellers must compare multiple options and decide which activities best match their requirements. A general list of popular attractions may not be appropriate for every user because preferences, available time, and spending limits differ.

Artificial intelligence can support travel planning by matching user preferences with relevant attractions and arranging them into a practical schedule. Rule-based artificial intelligence is suitable for a small travel-planning application because it uses predefined conditions to make decisions. For example, if a user selects history as an interest, the system can prioritize museums, monuments, and heritage sites. If the user's budget is limited, free or low-cost attractions can receive higher priority.

This paper presents an AI-Powered Personalized Travel Planning System that combines a travel knowledge base, a rule-based recommendation engine, an itinerary generation module, and a budget calculation module. The system produces a personalized day-wise plan and explains the reasons for its recommendations.

The main contributions of this work are:

- A rule-based process for matching user preferences with suitable attractions.
- A structured travel knowledge base containing attraction details and estimated expenses.
- A day-wise itinerary generation mechanism that considers trip duration and visit time.
- A budget estimation module that identifies potential spending-limit violations.
- An explainable recommendation process that does not depend on machine-learning or deep-learning training.

II. RELATED WORK

A. Recommendation Systems

Recommendation systems help users discover items or services that match their preferences. Adomavicius and Tuzhilin reviewed the development of recommender systems and discussed extensions to traditional recommendation approaches [16]. Ricci, Rokach, and Shapira presented a comprehensive overview of recommendation techniques and their applications [17]. These studies provide a foundation for understanding how user preferences can be matched with relevant choices.

In travel planning, recommendation systems can be used to identify attractions based on a user's interests, destination, budget, and available time. However, a useful travel planner must also consider practical constraints, such as the duration of each activity and the total cost of the itinerary.

B. Rule-Based Artificial Intelligence

Rule-based artificial intelligence represents knowledge using predefined conditions and actions. A typical rule follows the IF–THEN structure, where a condition is checked and an appropriate action is selected. Russell and Norvig describe knowledge-based reasoning as an important area of artificial intelligence [9].

For example, a travel system may apply the rule: *IF the user prefers history, THEN prioritize historical attractions.* Similarly, if a user specifies a low budget, the system can prioritize attractions with free admission or lower entry fees. These rules make the recommendation process transparent and relatively easy to maintain.

C. Personalized Tourist Itineraries

Research has examined recommendation rules for personalizing tourist itineraries in unfamiliar cities [3]. Personalized itinerary generation requires more than selecting individual attractions; it also requires arranging the selected activities into a schedule that respects the traveller's constraints.

The proposed project adopts a simple, explainable approach. Instead of training a machine-learning model, it uses explicit rules and structured attraction information. This makes the project suitable for an academic application where the recommendation logic must be understandable and testable.

D. Research Gap

Existing travel services may offer extensive information and sophisticated recommendation features, but a student-level project benefits from a transparent process that can be implemented and evaluated with limited resources. The proposed system focuses on connecting preference matching, daily scheduling, and estimated expenses in a single rule-based workflow. Its recommendations are limited by the coverage and accuracy of its knowledge base.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

Travellers often need to compare attractions, estimate costs, and organize activities across multiple sources. Generic travel lists may not reflect individual interests, available time, or budget. Manual planning can lead to an overloaded itinerary or an estimated cost that exceeds the user's spending limit.

The problem addressed in this project is to develop an intelligent travel-planning application that accepts user preferences and applies predefined rules to generate a personalized day-wise itinerary with an estimated budget.

B. Objectives

The objectives of the proposed system are:

1. To collect user requirements, including destination, duration, budget, number of travellers, and interests.
2. To maintain a structured knowledge base of tourist attractions and their details.
3. To recommend attractions using predefined rule-based conditions.
4. To arrange selected attractions into a day-wise itinerary.
5. To estimate expenses related to transportation, accommodation, food, and entry fees.
6. To identify possible budget overruns and suggest economical alternatives where available.
7. To provide understandable recommendations and evaluate the system using test cases.

IV. PROPOSED SYSTEM

A. System Overview

The proposed system receives travel preferences from the user and processes them through a sequence of modules. After validating the inputs, the system retrieves relevant attraction information from the travel knowledge base. The rule-based AI engine filters and ranks attractions according to user interests and budget. The itinerary generation module distributes the selected attractions across the requested number of days. Finally, the budget calculation module estimates the overall expense and checks whether it is within the specified budget.

The generated plan includes a day-wise list of activities, attraction details, estimated expenses, and explanations for selected recommendations.

B. User Input and Validation

The system collects the following information:

- **Destination:** The city or region the user wants to visit.
- **Trip duration:** The number of days available for the trip.
- **Budget:** The maximum estimated spending amount.
- **Number of travellers:** The number of people travelling.
- **Interests:** Nature, adventure, history, culture, shopping, or other supported categories.
- **Optional preferences:** A relaxed schedule, low-cost activities, or accommodation preferences.

The validation module checks that required fields are entered correctly. For example, trip duration and traveller count must be positive values, and the budget must be a valid non-negative amount.

C. Travel Knowledge Base

The knowledge base stores information about tourist attractions and estimated travel expenses. Each attraction record may include the attraction name, destination, category, estimated visit duration, entry fee, and approximate transportation cost. Accommodation and food estimates can be stored as separate records.

The knowledge base may be implemented using JSON, CSV, or a database. The system can use this information to identify attractions that match the user's preferences.

The quality of the recommendations depends on the completeness and accuracy of the stored information. Estimated prices must be treated as approximate values unless they are retrieved from a verified live data source.

D. Rule-Based Recommendation Engine

The rule-based AI engine applies explicit IF–THEN conditions to select suitable attractions. Examples of rules include:

- IF the selected interest is history, THEN prioritize museums, monuments, and heritage sites.

- IF the selected interest is nature, THEN prioritize parks, gardens, viewpoints, and nature attractions.
- IF the user has a limited budget, THEN rank free or low-cost attractions higher.
- IF an attraction's visit duration exceeds the available time, THEN move it to another day or exclude it from the current day's plan.
- IF the estimated trip cost exceeds the budget, THEN search for suitable lower-cost alternatives.

The system may assign a simple suitability score based on preference matches, estimated cost, and schedule compatibility. The score is calculated using predefined rules rather than a trained predictive model.

E. Itinerary Generation

After selecting suitable attractions, the itinerary generation module organizes them into a day-wise schedule. The module considers the total trip duration and the estimated time required for each activity.

For example, a three-day trip may have different attractions assigned to Day 1, Day 2, and Day 3. The system attempts to avoid placing too many activities within a single day. If distance information is available, nearby attractions can be grouped together to reduce unnecessary travel.

The generated itinerary is a suggested schedule. It should not be described as the fastest or shortest route unless a routing service has been integrated and tested.

F. Budget Calculation

The budget calculation module estimates the total trip expense by adding the main cost categories.

$$\textit{Estimated Total Cost} = \textit{Transportation} + \textit{Accommodation} + \textit{Food} + \textit{Entry Fees} + \textit{Other Expenses} \quad (1)$$

Per-person expenses can be multiplied by the number of travellers where appropriate. Accommodation can be estimated by room and number of nights, while food and entry fees can be calculated per person.

If the estimated total exceeds the user's budget, the system can display a warning and recommend lower-cost activities where suitable alternatives exist. The final estimate is intended to support planning rather than guarantee the actual amount spent.

V. SYSTEM ARCHITECTURE

The proposed system follows a modular architecture consisting of the user interface, input validation module, travel knowledge base, rule-based AI engine, itinerary generation module, budget calculation module, and result display module.

The user first enters travel preferences through the interface. The validation module checks the inputs and passes valid data to the recommendation process. The knowledge base supplies attraction and cost information to the rule engine. The itinerary generator arranges the selected attractions into a daily schedule, and the budget module calculates the estimated expenses. Finally, the result module displays the personalized travel plan.

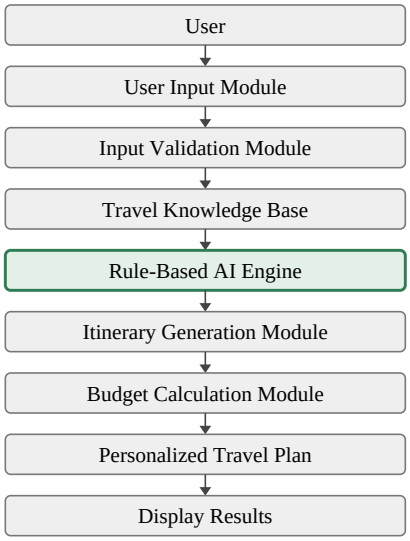

Fig. 1. Proposed architecture of the AI-Powered Personalized Travel Planning System. The rule-based AI engine (highlighted) is the decision-making component.

A. Module Descriptions

1. **User Interface:** Collects travel preferences and displays the generated itinerary.
2. **Input Validation Module:** Checks required fields and prevents invalid values from entering the recommendation process.
3. **Travel Knowledge Base:** Stores attraction details, categories, visit durations, and approximate costs.
4. **Rule-Based AI Engine:** Applies predefined conditions to filter and rank suitable attractions.
5. **Itinerary Generation Module:** Organizes selected attractions into a day-wise schedule based on the trip duration.
6. **Budget Calculation Module:** Calculates approximate expenses and checks them against the user's budget.
7. **Result Display Module:** Presents the itinerary, expense breakdown, budget warnings, and explanations for recommendations.

VI. IMPLEMENTATION METHODOLOGY

A. Processing Workflow

The system follows these steps:

1. The user enters the destination, trip duration, budget, number of travellers, and interests.
2. The input validation module checks the submitted information.
3. The system retrieves attraction records matching the selected destination.
4. The rule engine filters attractions according to interests and budget constraints.
5. Suitable attractions are ranked using predefined conditions.
6. The itinerary generator distributes the selected attractions across the available days.
7. The budget module estimates transportation, accommodation, food, and activity expenses.

8. The result module displays the day-wise itinerary, total estimated cost, and recommendation explanations.

B. Example Rule Execution

Consider a user who selects a destination, chooses history as an interest, requests a three-day itinerary, and enters a limited budget.

The rule engine identifies historical attractions in the knowledge base. It then prioritizes attractions that fit the user's spending limit. The itinerary generator assigns suitable attractions to each day while considering their estimated visit durations. The budget module calculates the estimated cost and checks it against the user's budget.

If the cost exceeds the limit, the system attempts to identify free or lower-cost alternatives from the available records.

C. Implementation Technologies

The system can be developed using a web interface, a programming language such as Python or JavaScript, and a JSON/CSV file or database for storing attraction information. The exact technology stack should reflect the implementation actually used in the project.

The core recommendation mechanism uses explicit rules and does not require machine-learning or deep-learning libraries.

VII. TESTING AND EVALUATION

The system should be evaluated using predefined test cases that cover recommendation quality, schedule constraints, budget calculation, and input validation.

TABLE I
TEST CASES FOR THE PROPOSED SYSTEM

Test ID	Test condition	Expected result
T1	User selects history as an interest	Historical attractions are prioritized.
T2	User selects nature as an interest	Nature-related attractions are recommended when available.
T3	User enters a limited budget	Free or low-cost activities receive higher priority.
T4	User selects a short trip duration	Activities are arranged within the configured daily time limit.
T5	Estimated expense exceeds the budget	A warning and suitable lower-cost alternatives are displayed where available.
T6	Required input is missing or invalid	The system asks the user to correct the input.
T7	No suitable attraction exists in the knowledge base	The system indicates that matching information is unavailable.
T8	User changes interests or budget	The system regenerates recommendations using the updated inputs.

A. Evaluation Measures

The system can be evaluated using the following measures:

- **Recommendation validity:** Whether the recommended attractions match the selected preferences.
- **Schedule validity:** Whether the total planned activity duration remains within the configured daily limit.
- **Budget calculation correctness:** Whether expense totals are calculated correctly from the stored values.
- **Rule traceability:** Whether each recommendation can be explained through an identifiable rule.
- **Input validation:** Whether invalid or incomplete inputs are handled appropriately.

These measures can be recorded during implementation testing. Actual test outcomes should be included only after the tests have been executed.

VIII. EXPECTED RESULTS AND DISCUSSION

The expected result is a personalized day-wise itinerary that reflects the destination, trip duration, budget, and interests provided by the user. The generated plan should organize suitable attractions into individual days and present an approximate expense breakdown.

The rule-based AI engine is expected to improve the organization of travel planning by reducing the need for users to manually compare every attraction. The budget module should help users recognize expensive activities and identify lower-cost alternatives. The explanation module should make the recommendations easier to understand because each selection can be linked to a predefined rule.

An important advantage of this approach is transparency. Unlike a recommendation model trained on historical user data, the proposed engine makes decisions using explicit conditions that can be reviewed and modified. This is useful for a small academic project where implementation simplicity and explainability are important.

However, the system's usefulness depends on the quality of the travel knowledge base. Missing attractions, outdated prices, and inaccurate visit durations may reduce the quality of the generated itinerary. The system should therefore communicate that its cost figures are estimates.

No measured performance results are claimed in this paper. The final version should include actual testing observations after the implementation has been evaluated.

IX. LIMITATIONS

The proposed system has several limitations. First, its recommendations are restricted to the attractions stored in the travel knowledge base. Second, transportation, food, accommodation, and entry-fee estimates may become outdated. Third, the system cannot guarantee current opening hours, ticket availability, weather, or traffic conditions without suitable live integrations.

The rule-based approach is transparent but depends on the completeness of its manually defined conditions. It may also require additional rules to handle complex preferences or unusual travel constraints. Finally, the generated itinerary is a suggested schedule and does not guarantee the most efficient travel route.

These limitations can be addressed gradually by expanding the knowledge base, validating cost information, improving scheduling rules, and integrating reliable external services.

X. CONCLUSION AND FUTURE WORK

This paper presented the design of an AI-Powered Personalized Travel Planning System that uses rule-based artificial intelligence to generate customized itineraries and estimate travel expenses. The proposed system combines user preferences, a structured travel knowledge base, explicit recommendation rules, day-wise scheduling, and budget calculation to make travel planning more organized and understandable.

The main benefit of the proposed approach is that its decisions can be explained using predefined rules. It is suitable for a student-level project because it does not require machine-learning or deep-learning training. The system can be tested using clear test cases for recommendation validity, schedule constraints, budget calculations, and input validation.

Future work may include integrating map and routing services, live weather information, attraction opening hours, and updated ticket and accommodation prices. Additional enhancements may include multilingual support, downloadable PDF itineraries, saved travel plans, group-travel preferences, accessibility options, and a mobile-friendly interface. Further evaluation should measure the correctness of itinerary generation, budget calculations, and user feedback.

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